Pooling Layers

Deep Learning

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Contents of This Video

In this video, we will cover:

- Why we need pooling layers
- Max pooling operation
- Average pooling operation
- How pooling provides invariance
- Pooling parameters and output size

Why Pool?

Convolution preserves spatial size (with same padding)

- 224×224 input → 224×224 feature map
- Too large for deep networks!

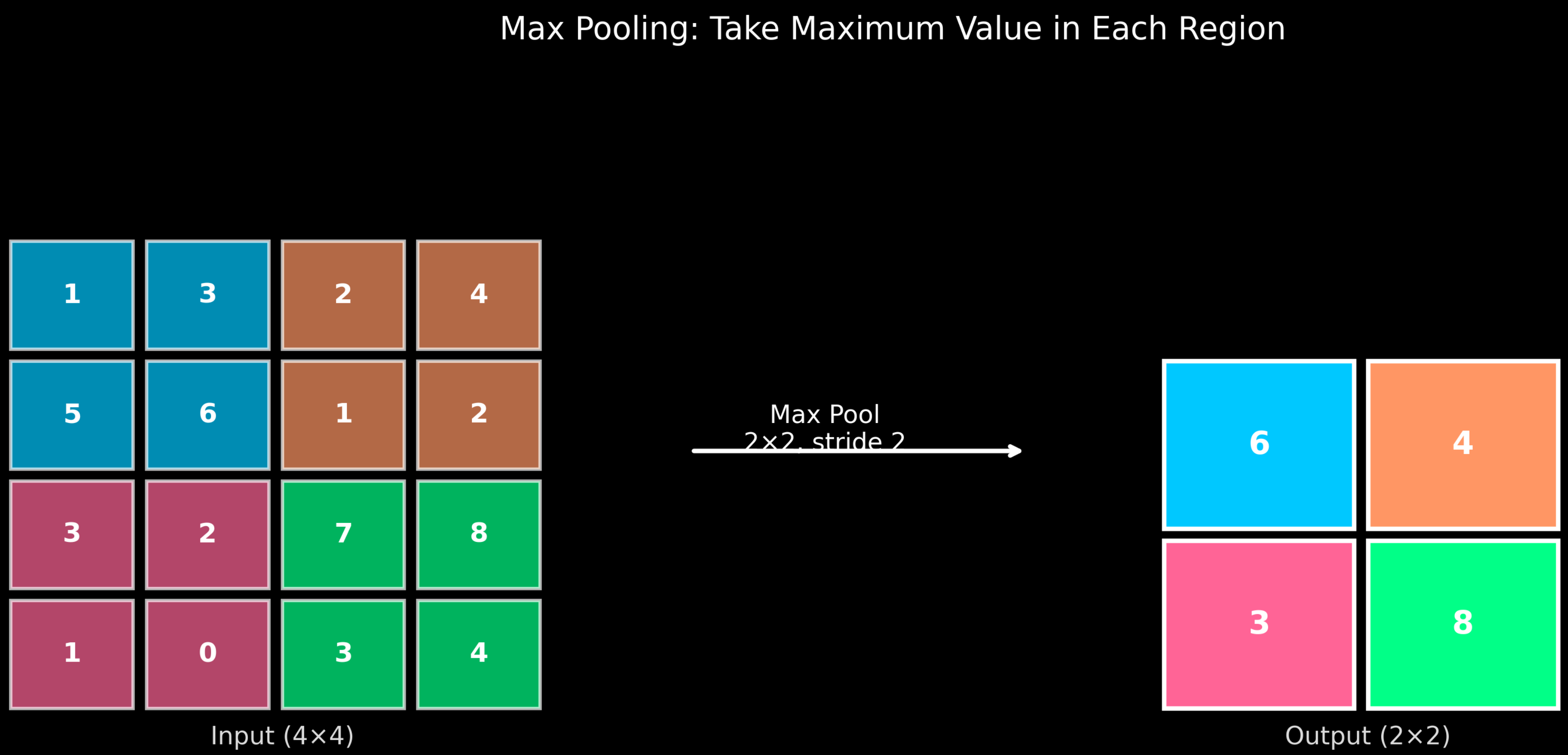
Pooling reduces spatial dimensions:

- 224×224 → 112×112 → 56×56 → 28×28 → 14×14 → 7×7

Benefits:

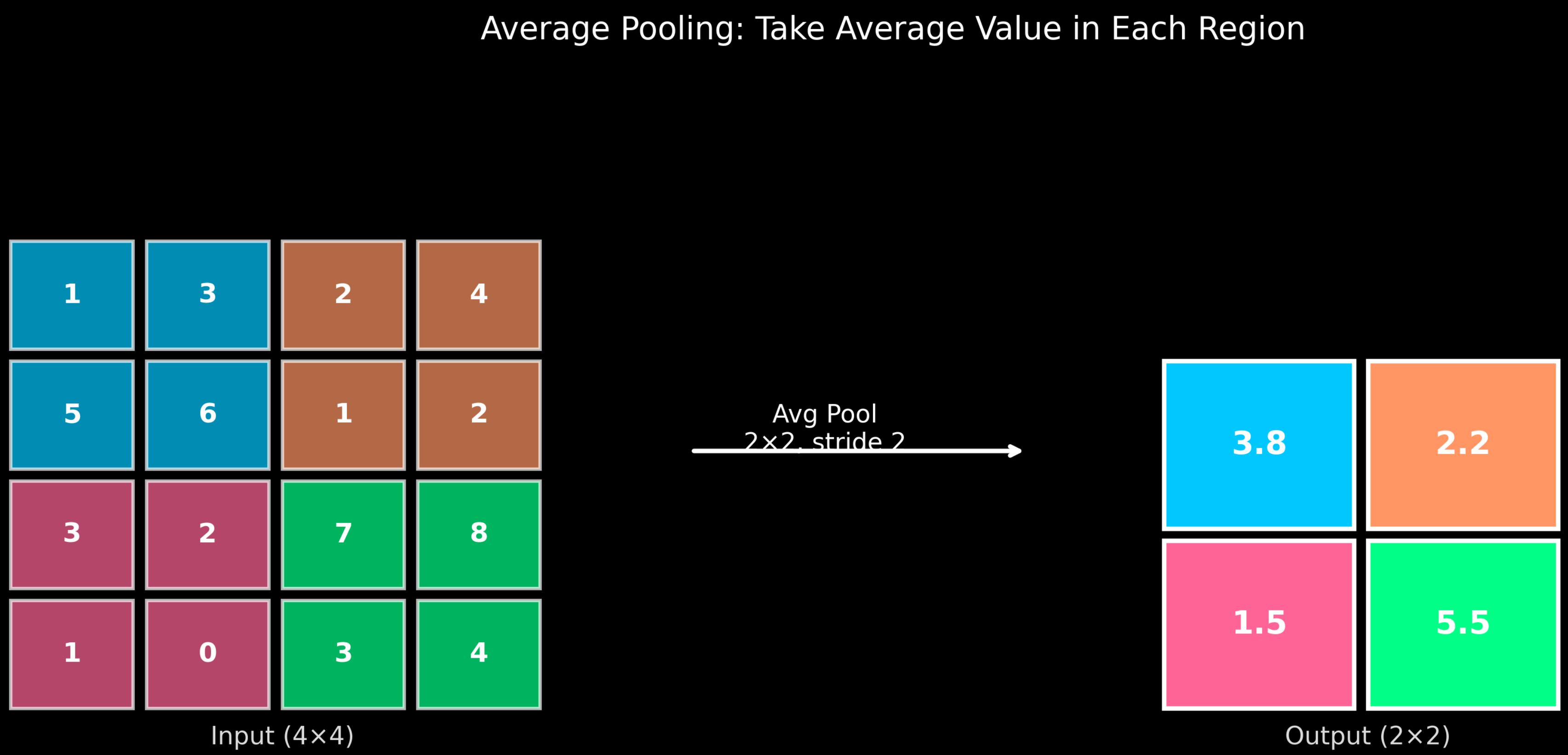
- Fewer parameters in later layers
- Faster computation
- Larger receptive field
- Translation invariance

Max Pooling



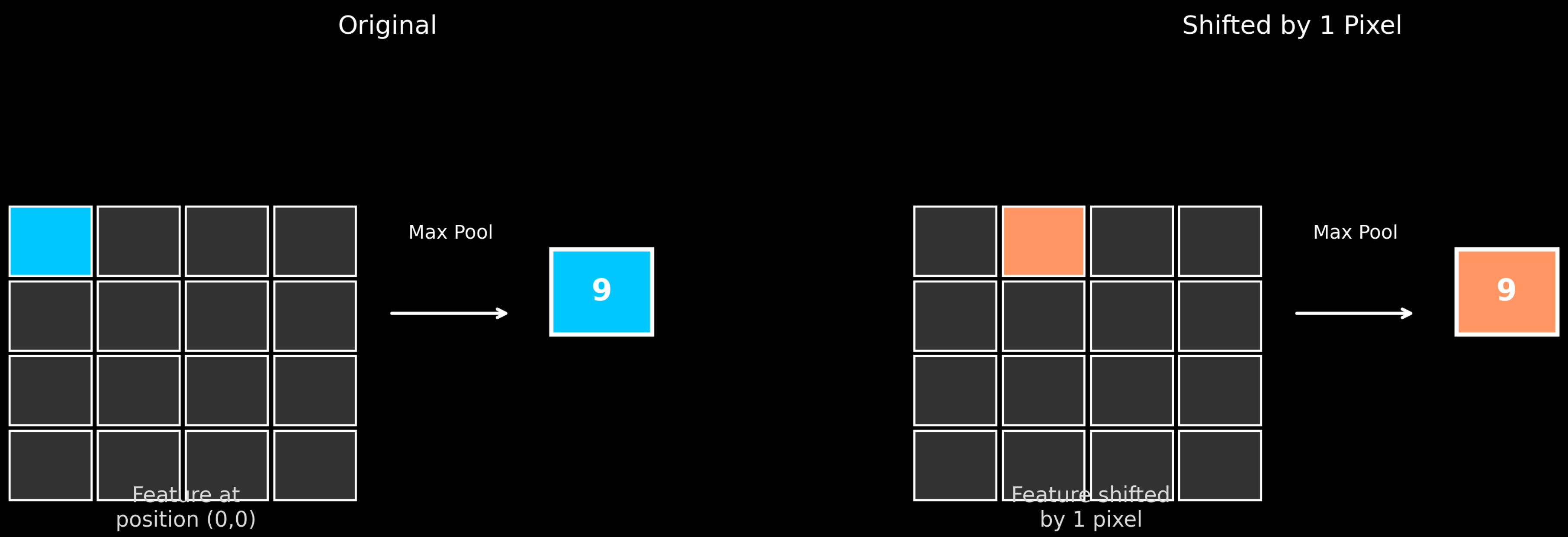
Max Pooling: Take Maximum Value in Each Region
1
3
2
4
5
6
1
2
3
2
7
8
1
0
3
4
Input (4×4)
Max Pool
2×2, stride 2
6
4
3
8
Output (2×2)

Average Pooling



Translation Invariance

Max pooling makes the output robust to small shifts:



Same output despite the shift!

Common Pooling Settings

Setting	Value	Effect
Pool size	2×2	Size of pooling window
Stride	2	Usually equals pool size
Type	Max	Most common for CNNs

Global Average Pooling:

- Pool size = entire feature map
- 7×7×512 → 1×1×512 (just 512 numbers)
- Often used before final classification layer

What We've Covered

In this video, we've learned:

- Pooling reduces spatial dimensions of feature maps
- Max pooling: take maximum value in each window
- Average pooling: take average value in each window
- Pooling provides translation invariance
- Common setting: 2×2 max pooling with stride 2
- Global average pooling at network end